

Pricing Caplets: Black and Bachelier

6.1 Setting up the pricing problem

A caplet pays $N\tau \max(F_T - K, 0)$ at (or discounted back from) time T_2 , where $F_T = L(T_1; T_1, T_2)$ is the random forward rate observed at the fixing date T_1 . To price it, we need the expected value of this payoff under the appropriate risk-neutral measure, discounted back to today. Chapter 12 will make the choice of measure completely rigorous; for now, the key fact — proved properly later, used freely now — is that under the T_2 -**forward measure** $\mathbb{Q}^{T_2}$ (the measure associated with using $P(\cdot, T_2)$ as numeraire), the forward rate $F_t = L(t; T_1, T_2)$ is a *martingale with no drift*, and the caplet price collapses to

$$\text{Caplet}(0) = N\tau P(0, T_2) \mathbb{E}^{\mathbb{Q}^{T_2}} [\max(F_{T_1} - K, 0)].$$

Everything in this chapter is a choice of *what distribution* to assume for F_{T_1} under this measure, followed by evaluating the expectation in closed form.

6.2 Black's formula: the lognormal choice

Assume F_t follows driftless geometric Brownian motion under $\mathbb{Q}^{T_2}$: $dF_t = \sigma F_t dW_t$, with constant σ . This is exactly the Black–Scholes assumption, applied to a forward rate instead of a forward stock price, and it is historically where Fischer Black's 1976 paper on pricing options on futures/forwards was first applied to caps. The resulting formula was, for decades, the default in rates markets.

Key result: Black's formula for a caplet

$$\text{Caplet}_{\text{Black}}(0) = N\tau P(0, T_2) [F \Phi(d_1) - K \Phi(d_2)]$$

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T_1}{\sigma\sqrt{T_1}}, \quad d_2 = d_1 - \sigma\sqrt{T_1}$$

where $F = L(0; T_1, T_2)$ is today's forward rate, σ is the lognormal (Black) volatility, T_1 is the time to fixing, and $\Phi(\cdot)$ is the standard normal CDF. The corresponding floorlet is $N\tau P(0, T_2) [K\Phi(-d_2) - F\Phi(-d_1)]$.

6.3 Bachelier's formula: the normal choice

Assume instead $dF_t = \sigma_n dW_t$ under $\mathbb{Q}^{T_2}$: an absolute, arithmetic random walk with no drift. This is the older of the two formulas historically (Louis Bachelier's 1900 thesis predates Black and Scholes by nearly seventy years) but the more recently market-dominant, precisely because — as Chapter 1 established — it handles a near-zero or negative forward rate without difficulty.

Key result: Bachelier's formula for a caplet

$$\text{Caplet}_{\text{Bachelier}}(0) = N\tau P(0, T_2) \sigma_n \sqrt{T_1} \left[d \Phi(d) + \varphi(d) \right], \quad d = \frac{F - K}{\sigma_n \sqrt{T_1}}$$

where $\varphi(\cdot)$ is the standard normal density. At the money ($F = K$, so $d = 0$), this collapses to the strikingly simple

$$\text{Caplet}_{\text{Bachelier}}^{\text{ATM}}(0) = N\tau P(0, T_2) \sigma_n \sqrt{\frac{T_1}{2\pi}}$$

a formula simple enough to compute in one's head to a good approximation, and the source of the mental-arithmetic trick many rates traders use to sanity-check a quoted price against a bp-vol quote.

Worked example 6.1 — pricing the 4.10% caplet from Example 5.1 properly.

Recall the third caplet from Worked Example 5.1: notional \$10mm, strike $K = 4.00\%$, forward $F = 4.10\%$, fixing $T_1 = 1.5$ years, $\tau = 0.25$, discount factor $P(0, T_2) = 0.9702$. Suppose the market quotes a normal volatility of $\sigma_n = 90$ bp/yr for this caplet. Price it with Bachelier's formula.

Step 1 — compute d .

$$d = \frac{F - K}{\sigma_n \sqrt{T_1}} = \frac{0.0010}{0.0090 \times \sqrt{1.5}} = \frac{0.0010}{0.011022} = 0.09073$$

Step 2 — evaluate $\Phi(d)$ and $\varphi(d)$.

$$\Phi(0.0907) \approx 0.5361, \quad \varphi(0.0907) \approx 0.39734$$

Step 3 — assemble the bracket.

$$d \Phi(d) + \varphi(d) = 0.09073 \times 0.5361 + 0.39734 = 0.04864 + 0.39734 = 0.44598$$

Step 4 — apply the formula.

$$\begin{aligned} \text{Caplet} &= \$10\text{mm} \times 0.25 \times 0.9702 \times 0.0090 \times \sqrt{1.5} \times 0.44598 \\ &= \$10\text{mm} \times 0.25 \times 0.9702 \times 0.011022 \times 0.44598 \approx \$11,920 \end{aligned}$$

Sanity check. The discounted intrinsic value of this caplet, from Worked Example 5.1, was $\$2,500 \times 0.9702 \approx \$2,426$. The full Bachelier price of \$11,920 is comfortably above that, exactly as it must be: an option is always worth at least its intrinsic value, and the gap is the time value contributed by the roughly 90 bp of uncertainty in where the rate actually fixes in 1.5 years.

Interpretation. The time value here — about \$9,500, the difference between the full price and the discounted intrinsic value — is large relative to intrinsic value precisely because this caplet is only marginally in the money at the forward (10 bp): its value comes almost entirely from the chance that the rate moves further into the money before fixing, not from where it happens to sit today. The same arithmetic applied to the first two, currently out-of-the-money, periods of Example 5.1 would show an even starker contrast: zero intrinsic value by construction, yet strictly positive full value once volatility is correctly priced in.

PITFALL

Black and Bachelier volatilities are not interchangeable numbers and cannot be swapped into the wrong formula, nor can a Black vol from one source be silently averaged with a Bachelier vol from another without converting first (Chapter 1’s approximate conversion, or better, matching prices exactly via each formula, both work — but skipping the conversion entirely does not). A second, subtler trap: Black’s formula is *undefined*, not merely inaccurate, for $F \leq 0$, because $\ln(F/K)$ requires $F > 0$. Any system that might encounter a negative forward rate — which did happen in EUR and JPY markets for years, and remains possible again — must either use Bachelier throughout, or use a *displaced* or *shifted* lognormal formula (Chapter 15) that adds a constant shift before taking the logarithm.

6.4 Cap–floor parity, checked

The identity from Chapter 5 is easy to confirm algebraically for either formula: subtracting the floorlet formula from the caplet formula and simplifying (using $\Phi(d_1) - 1 = -\Phi(-d_1)$, etc.) collapses every volatility-dependent term and leaves exactly $N\tau P(0, T_2)(F - K)$, independent of σ or σ_n entirely. This is a genuinely useful implementation check: if a caplet and floorlet pricing function, evaluated at the same volatility, don’t satisfy parity to machine precision, there is a bug in the code, not a modelling approximation.

ON THE DESK

In practice, a cap or floor is not quoted with a separate volatility per caplet. The market convention (both historically for LIBOR and today for SOFR) is to quote a single **flat volatility** per cap tenor (e.g. “the 5-year cap trades at 85 normal”), applied uniformly across every caplet in that cap regardless of each caplet’s individual fixing date. Recovering the volatility that applies to each *individual* caplet — the **caplet** or **spot** volatility — from a ladder of flat cap quotes is a genuine bootstrapping problem in its own right, solved by a vega-weighted stripping procedure. This is one of the two calibration inputs (the other being swaptions) that Chapter 16 builds in full, and it is usually the very first step any BGM calibration pipeline runs.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. Write Black’s formula and Bachelier’s formula for a caplet, and identify d_1 , d_2 (Black) and d (Bachelier).
2. Price a caplet numerically given a forward rate, strike, vol, fixing time and discount factor, using either formula.
3. Explain why Black’s formula fails structurally for a negative forward rate, and why Bachelier’s does not.

